Functions of Several Variables

Definition (Multivariate Functions): A function f of n real variables is a rule which assigns a unique real number $f(x_1, x_2, \dots, x_n)$ to each point $(x_1, x_2, \dots x_n)$ in some subset $D(f)$ of $\mathbb{R}^n$.

Like with single variable functions, multivariate functions have both a domain, $D(f)$, and a range (which is just obtained from points in the domain).

A simple and well-known function is the function $f(x,y) = x^2 + y^2 = r$.

Definition (Multivariate Limit): We say that $\lim_{(x,y)\to(a,b)} f(x,y) = L$ iff for every positive number ϵ there exists a positive number $\delta = \delta(\epsilon)$, depending on ϵ, such that

$$|f(x,y) - L| < \epsilon \quad \text{whenever} \quad 0 < \sqrt{(x-a)^2 + (x-b)^2} < \delta.$$

Note that many of the simpler single-variable limit rules apply here too. So for example, $\lim_{(x,y)\to(a,b)} \frac{f(x,y)}{g(x,y)} = \frac{L}{M}$.

Definition (Multivariate Continuity): The function $f(x,y)$ is continuous at the point (a,b) if

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b).$$

Partial Derivatives

Definition (Partial Derivative): The partial derivatives of the function $f(x,y)$ with to the variables x and y are the functions f_x and f_y given by

$$f_x = \lim_{\Delta \to 0} \frac{f(x+\Delta, y) - f(x,y)}{\Delta}$$

$$f_y = \lim_{\Delta \to 0} \frac{f(x, y+\Delta) - f(x,y)}{\Delta},$$

provided the limits exist.

For single-variable derivatives, we normally write $\frac{d}{dx}[f(x)]$ or $\frac{dy}{dx}$. To denote the partial derivative, we will write $\frac{\partial}{\partial x}[f(x,y)]$ or $\frac{\partial z}{\partial x}$. The same works for y: $\frac{\partial z}{\partial y} = \frac{\partial}{\partial y}[f(x,y)] = f_y$. When read aloud, we would say "partial z with respect to x" or the equivalent for y.

Definition (Values of Partial Derivatives):

$$\left.\frac{\partial z}{\partial x}\right|_{(a,b)} = \left.\left(\frac{\partial}{\partial x} f(x,y)\right)\right|_{(a,b)} = f_x(a,b)$$

$$\left.\frac{\partial z}{\partial x}\right|_{(a,b)} = \left.\left(\frac{\partial}{\partial y} f(x,y)\right)\right|_{(a,b)} = f_y(a,b).$$

Definition (Pure second partial derivative): $\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}\frac{\partial z}{\partial x} = f_{xx}$ and $\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y}\frac{\partial z}{\partial y} = f_{yy}$.

Definition (Mixed second partial derivative): $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x}\frac{\partial z}{\partial y} = f_{yx}$ and $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y}\frac{\partial z}{\partial x} = f_{xy}$.

Theorem (Chain Rule I): If z is a function of x and y with continuous first partial derivatives, and if x and y are differential functions of t, then

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}.$$

Theorem (Chain Rule II): $$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}, \quad \text{where} \quad \begin{pmatrix} \frac{\partial z}{\partial s} & \frac{\partial z}{\partial t} \end{pmatrix} = \begin{pmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \end{pmatrix} \begin{pmatrix} \frac{\partial x}{\partial s} & \frac{\partial x}{\partial t} \\ \frac{\partial y}{\partial s} & \frac{\partial y}{\partial t} \end{pmatrix}.$$

Definition (Differentiability): We say that the function $f(x,y)$ is differentiable at the point (a,b) if

$$\lim_{h,k\to 0} \frac{f(a+h,b+k) - f(a,b) - hf_1(a,b) - f_2(a,b)}{\sqrt{h^2+k^2}} = 0.$$

Definition (Level Sets): Given a function f of n variables and a constant c in the range of f, the set of points in the domain of f with $f=c$ is called a level set.

Gradient and Directional Derivatives

Definition (Gradient Vector): At any point (x,y) where the first partial derivatives of the function $f(x,y)$ exist we define the gradient vector $\nabla f(x,y)$ by

$$\nabla f(x,y) = f_x(x,y)\mathbf{i} + f_y(x,y)\mathbf{j} = \begin{bmatrix} f_x(x,y) \\ f_y(x,y) \end{bmatrix}.$$

For instance, if $f(x,y) = x^2+y^2$, then $\nabla f(x,y) = 2x\mathbf{i} + 2y\mathbf{j} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$.

Theorem (Normal vector through gradient vector): If $f(x,y)$ is differentiable at the point (a,b) and $\nabla f(a,b) \neq 0$, then $\nabla f(a,b)$ is a normal vector to the level curve of f which passes through (a,b).

Definition (Directional derivative): Let r be a unit vector, that is,

$$r = ui + vj \quad \text{where } u^2 + v^2 = 1.$$

The directional derivative of $f(x,y)$ at (a,b) in the direction of r is the rate of change of $f(x,y)$ with respect to distance measured at (a,b) along a line in the direction of r in the plane xy-plane. This directional derivative is given by

$$D_r f(a,b) = \lim_{\Delta \to 0} \frac{f(a+\Delta u, b+\Delta v) - f(a,b)}{\Delta},$$

or, equivalently,

$$D_r f(a,b) = \frac{d}{dt} f(a+tu, b+tv)\Big|_{t=0}.$$

Theorem (Formula for the directional derivative): $D_r f(a,b) = r \cdot \nabla f(a,b).$

Definition (Jacobian Determinants): The Jacobian of two functions, $u = u(x,y)$ and $v = v(x,y)$, with respect to two variables, x and y, is the determinant

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial u}{\partial y} \end{vmatrix}.$$

Similarly, the Jacobian of two functions, $F(x,y)$ and $G(x,y)$, with respect to the variables, x and y, is the determinant

$$\frac{\partial(F,G)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial F}{\partial x} & \frac{\partial F}{\partial y} \\ \frac{\partial G}{\partial x} & \frac{\partial F}{\partial y} \end{vmatrix} = \begin{vmatrix} F_x & F_y \\ G_x & G_y \end{vmatrix}$$

Theorem (Extreme values): A function $f(x,y)$ can have a local or absolute extreme value at a point (a,b) in its domain only if (a,b) is either

1. A critical point: $\nabla f(a,b) = 0$
2. A singular point of: $\nabla f(a,b)$ does not exist
3. A boundary point of the domain of f.

Theorem: If f is a continuous function of n variables whose domain is a closed and bounded set in $\mathbb{R}^n$, then the range of f is a bounded set of real numbers, and there are points in its domain where f takes on absolute maximum and minimum values.

Theorem (Second derivative test): Suppose (a,b) is a critical point of $f(x,y)$ interior to the domain of f. Suppose also that the second partial derivatives of f are continuous near (a,b) and have at that point the values

$$A = f_{xx}(a,b), \quad B = f_{xy}(a,b) = f_{yx}(a,b), \quad C = f_{yy}(a,b).$$

If $B^2 < AC$ and $A > 0$ then f has a local minimum value at (a,b).
If $B^2 < AC$ and $A < 0$ then f has a local maximum value at (a,b).
If $B^2 > AC$ then f has a saddle point at (a,b).

Note: similar with ratio tests, $B^2 = AC$ provides no information

If $r = ui + vj$ is a unit vector then the second directional derivative of $f(x,y)$ at (a,b) in the direction of r is given by

$$D_r^2 f(a,b) = r \cdot \nabla (r \cdot \nabla f)(a,b)$$
$$= Au^2 + 2Buv + Cv^2.$$

As it is a homogeneous polynomial of degree two, the expression $Q(u,v) = Au^2 + 2Buv + Cv^2$ is called a quadratic form in the variables u and v. Its said to be positive definite if $Q(u,v) > 0$ for all nonzero vectors r (likewise, negative definite $\Leftarrow Q(u,v) < 0$). Otherwise, it is said to be indefinite.

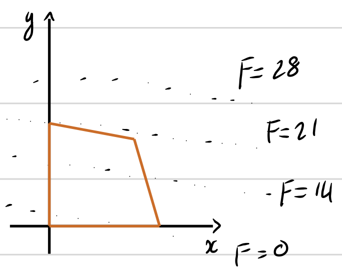
Linear Programming

Linear programming is a branch of linear algebra which develops systematic techniques for min / max value problems for linear functions subject to several linear inequality constraints. An example is $ax + by \leq c$ with a solution set consisting of a half-plane lying on one side of $ax + by = c$.

Find the maximum value of $F(x,y) = 2x + 7y$ subject to the constraints

$$x + 2y \leq 6,$$
$$2x + y \leq 6,$$
$$x \geq 0,$$
$$y \geq 0.$$

Solution: The solution set S of the system is the quadrilateral region with vertices $(0,0)$, $(3,0)$, $(2,2)$, and $(0,3)$.



Here, we have several level curves of the linear function F, parallel to the function in question $(-\frac{2}{7})$. The line that gives F the greatest value and still intersects S passes through $(0,3)$. Thus, the maximum value of F subject to constraints is 21.

It is often very difficult or impossible to solve the constraint equations, so we need a new technique. Lets say we want to minimize or maximise $f(x,y)$ subject to $g(x,y) = 0$. Suppose there exists a point $P_0 = (x_0, y_0)$ on the curve C with equation $g(x,y) = 0$, such that $f(x,y)$, restricted to C, has an extreme value at P_0. We also assume that

1. P_0 is not an endpoint of C
2. $\nabla g(P_0) \neq 0$.

After "a lot" of assumptions, we reach the idea that we should look for the critical points of the lagrangian function

$$L(x,y,\lambda) = f(x,y) + \lambda g(x,y).$$

Any critical point of L must have

$$\left.\begin{aligned} 0 &= \frac{\partial L}{\partial x} = f_x(x,y) + \lambda g_x(x,y) \\ 0 &= \frac{\partial L}{\partial y} = f_y(x,y) + \lambda g_y(x,y) \\ 0 &= \frac{\partial L}{\partial \lambda} = g(x,y) \end{aligned}\right\} \text{ basically, } \nabla f \text{ is parallel to } \nabla g$$

Taylor Series

For single-variable functions, we defined the taylor series for some function $f(x)$ as $\sum_{n=0}^{\infty} \frac{x^n \cdot \frac{d^n y}{dx^n}}{n!}$, or using the lagrange form of the remainder, $\sum_{n=0}^{m} \frac{x^n \cdot \frac{d^n y}{dx^n}}{n!} + \frac{\frac{d^{m+1} y}{dx^{m+1}}\big|_\theta}{(m+1)!}$.

Definition (Heissian matrix): The Hessian matrix, or 'Hessian', of $f(x,y)$ is

$$\text{Hess } f = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}.$$

Theorem (Taylor approximation): Let $f(x,y)$ have continuous partial derivatives of order up to 2, and let the point (a,b) be in its domain. Then the first and second Taylor approximations of f at (a,b)

$$T^{(1)}_{(a,b)} f(x,y) = f(a,b) + \nabla f(a,b)\begin{bmatrix} x-a \\ y-b \end{bmatrix}$$

$$T^{(2)}_{(a,b)} f(x,y) = f(a,b) + \nabla f(a,b)\begin{bmatrix} x-a \\ y-b \end{bmatrix} + \frac{1}{2}\begin{bmatrix} x-a & y-b \end{bmatrix} \text{Hess } f(a,b) \begin{bmatrix} x-a \\ y-b \end{bmatrix}$$

where we omit the subscript (a,b) if the point, from which the approximation is carried out, is clear. Close to that point, we have

$$T^{(1)}f \approx f, \quad T^{(2)}f \approx f,$$

where the second-order approximation is better in the sense that the error $|T^{(2)}f(x,y) - f(x,y)|$ decays faster than $|T^{(1)}f(x,y) - f(x,y)|$ as $(x,y) \to (a,b)$. Similarly for $f = f(x,y,z)$.

If a more 'rigerous-looking' definition of the Taylor formula itself, we have

$$\begin{aligned} f(a+h, b+k) &= \sum_{m=0}^{n} \frac{1}{m!} (h D_x + k D_y)^m f(a,b) + R_n(h,k) \\ &= \sum_{m=0}^{n} \sum_{j=0}^{m} C_{mj} D_x^j D_y^{m-j} f(a,b) h^j k^{m-j} + R_n(h,k), \end{aligned}$$

where, using binomial expansion, we have

$$C_{mj} = \frac{1}{m!} \binom{m}{j} = \frac{1}{j!(m-j)!},$$

and where the remainder term is given by

$$R_n(h,k) = \frac{1}{(n+1)!} (h D_x + k D_y)^{n+1} f(a+\theta h, b+\theta k)$$

for some θ between 0 and 1. If f has partial derivatives of all orders and $\lim_{n\to\infty} R_n(h,k) = 0$, then $f(a+h, b+k)$ can be expressed as a Taylor series in powers of h and k:

$$f(a+h, b+k) = \sum_{m=0}^{\infty} \sum_{j=0}^{m} \frac{1}{j!(m-j)!} D_x^j D_y^{m-j} f(a,b)(x-a)^j (y-b)^{m-j}.$$

As for functions of one variable, the Taylor polynomial of degree n:

$$P_n(x,y) = \sum_{m=0}^{n} \sum_{j=0}^{m} \frac{1}{j!(m-j)!} D_x^j D_y^{m-j} f(a,b)(x-a)^j (y-b)^{m-j},$$

provides the "best" nth degree polynomial approximation for $f(x,y)$ near (a,b). For $n=1$, this approximation reduces to the tangent plane approximation

$$f(x,y) \approx f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b).$$